

Homework Set 8 - DAACS

Purpose: The purpose of this assignment is to practice constructing a relational database and its hard metadata. In order to know what data you must collect (and in what form) you must have a goal in mind as well as an analysis plan. To that end we provide you with sample analysis code for the first part of this homework and you have to make the data it requires. Then in the second part you are on your own and continuing to work on your project. For that part you will have to make everything yourself.

Instructions: Put your solutions for all items in a google collaborative notebook. Create a level-1 header for each section (new dataset for the week and project data). Within each section use level-2 headers for the item groups and level-3 headers for the individual items. If items indicate a header structure then that overrides these general instructions. The submission is made via a link on UVACanvas.

Data

You will be generating your dataset by querying from the DAACS project website <https://www.daacs.org/query-the-database/>.

This dataset will be created to meet the needs of the sample analysis code here https://github.com/DAACS-Research-Consortium/DAACS-Open-Academy/blob/main/FSS2021/Workshop1/RWorkshop1_SCV.R.

8.1 DAACS

For this section all items are based on the DAACS dataset you create to meet the needs of the given sample code. In addition you will note several hints/notes. This section requires considerable judgement on your part, and learning how to apply that judgement is a key part of this assignment.

Hint: The correct dataset for this section has 1,705 records in a single table.

Data Creation (10 ψ)

(2 ψ) **Item 1. Paragraph** or two explaining the raw data acquisition process (provenance). (**Hint:** Explain what you did.)

(3 ψ) **Item 2. Code** For this dataset there is no code, enjoy your 3 free points. Put this item in your solution with the text "3 free points". Table showing the code used to create the data, one row per file, with a brief description and link to code in repo

(1 ψ) **Item 3. Bias Identification** description of how bias could be/was introduced in the data collection process. (In this case answer based on information from the DAACS website about how the DAACS data was entered into the database).

(1 ψ) **Item 4. Bias Mitigation** description of how biases can be handled/quantified/accounted for in analysis. (Same as before, this is about DAACS data, not your secondary data).

(3 ψ) **Item 5. Rationale** for critical decisions, especially judgement calls, and places that can introduce/mitigate uncertainty

Metadata (10 ψ)

(3 ψ) **Item 1. Schema** ER diagram at the logical level. (This will only have one entity, don't forget to indicate the primary key).

(1 ψ) **Item 2. Data** Table listing all of the tables in the dataset, one line per table, brief description and link to csv file

(3 ψ) **Item 3. Data Dictionary** Table with one row per feature in the data set containing: name, data type, description, example

(3 ψ) **Item 4. Data Dictionary** quantification of uncertainty for numerical features

Pipeline Check (5 ψ)

Remember, best practice is to check that your work actually works.

(2 ψ) **Item 1. Demonstrate** that the sample code runs using your the data you created.

(3 ψ) **Item 2.** The sample code and your data produce the correct answer.

8.2 Project 1

For this section all items are based on your specific project.

Data Creation (10 ψ)

(2 ψ) **Item 1. Paragraph** or two explaining the raw data acquisition process (provenance).

(3 ψ) **Item 2. Code** Table showing the code used to create the data, one row per file, with a brief description and link to code in repo

(1 ψ) **Item 3. Bias Identification** description of how bias could be/was introduced in the data collection process

(1 ψ) **Item 4. Bias Mitigation** description of how biases can be handled/quantified/accounted for in analysis

(3 ψ) **Item 5. Rationale** for critical decisions, especially judgement calls, and places that can introduce/mitigate uncertainty

Metadata (10 ψ)

(3 ψ) **Item 1. Schema** ER diagram at the logical level

(1 ψ) **Item 2. Data** Table listing all of the tables in the dataset, one line per table, brief description and link to csv file

(3 ψ) **Item 3. Data Dictionary** Table with one row per feature in the data set containing: name, data type, description, example

(3 ψ) **Item 4. Data Dictionary** quantification of uncertainty for numerical features

Pipeline Check (5 ψ)

Remember, best practice is to check that your work actually works.

(2 ψ) **Item 1. Demonstrate** that the sample code runs using your the data you created. (**N.B.** This is not your final analysis code. This can be shorter proof of concept code. Trivial solutions earn no credit for this item and the following item.)

(3 ψ) **Item 2.** The sample code and your data produce the correct answer.